Variational and Optimization-based Approaches for Event-Based Vision Applications

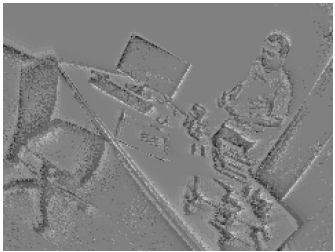
From unobserved modes to well-posed estimators

Daniel Blauvelt

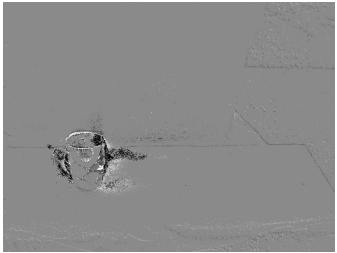
Department of Mathematics ▪ George Mason University
Dissertation Director: Harbir Antil

Dissertation Defense ▪ Fall 2026

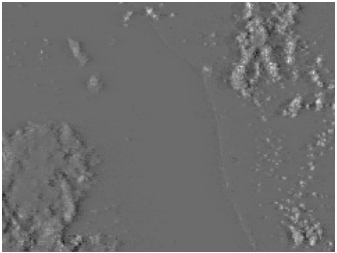
Events alone recover a handheld office, a bullet hitting a cup, and Earth from orbit



Office, handheld



Bullet hits a cup

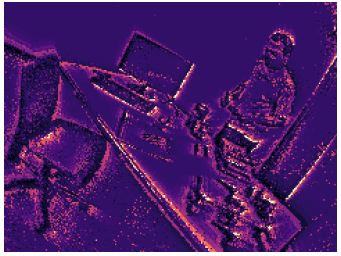


Earth from the ISS (Falcon Neuro)

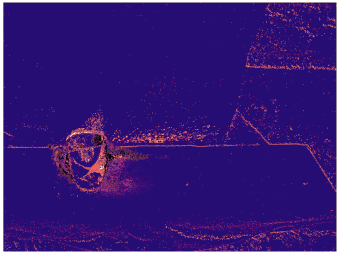
What can brightness changes reveal about images, motion, and depth—and at what computational cost?

Dissertation: Figure 2.7, Figure 2.8, and Figure 2.9; Section 2.3.3. Data: office [36], cup [37], Falcon Neuro [38].

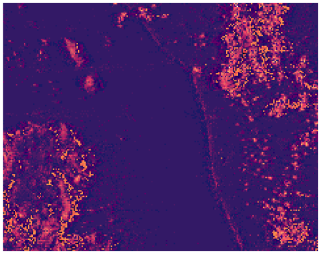
Events alone recover a handheld office, a bullet hitting a cup, and Earth from orbit



Office, handheld



Bullet hits a cup

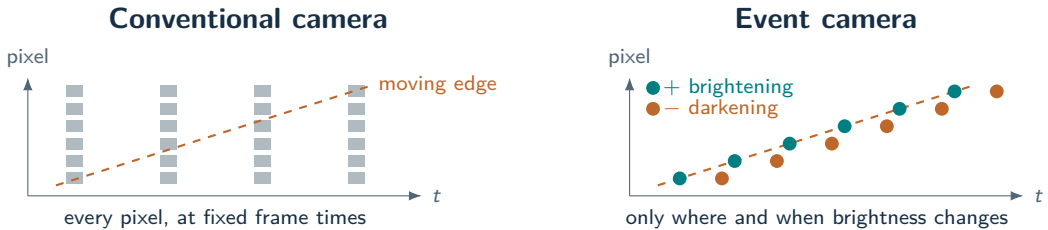


Earth from the ISS (Falcon Neuro)

What can brightness changes reveal about images, motion, and depth—and at what computational cost?

Dissertation: Figure 2.7, Figure 2.8, and Figure 2.9, false-color renderings of the same reconstructed frames.

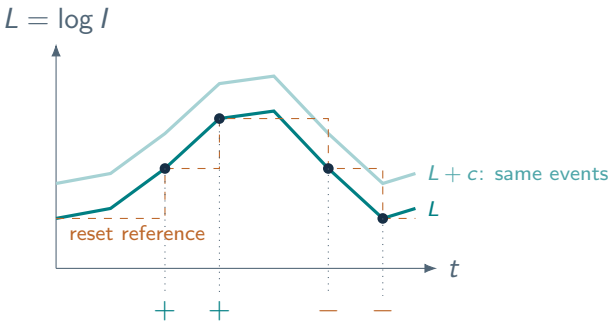
Why an event camera? Each pixel reports its own changes, with microsecond timing and 140 dB range



- Microsecond timestamps, about 140 dB dynamic range versus 60 dB, low power, little motion blur [1].

Dissertation: Chapter 2, neuromorphic camera overview; [1]. Diagram is schematic.

Each event marks one threshold of log-brightness change, so the baseline level stays free



An event is a tuple

$$e_k = (\mathbf{x}_k, s_k, p_k), \qquad p_k = \pm 1.$$

For the ideal threshold model,

$$L(\mathbf{x}_k, s_k) - L(\mathbf{x}_k, s_k^-) = p_k c_{\mathrm{th}}.$$

Dissertation: Eq. 2.1; Eq. 3.3; ideal threshold model [20]. Diagram is schematic.

From events to images, motion, and depth: each estimate has its own unobserved quantity

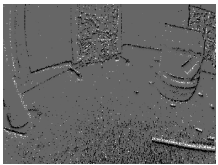
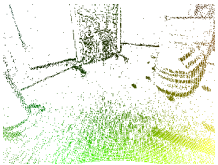


Image reconstruction (Ch. 2)

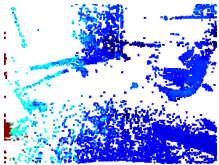
Unobserved: the brightness baseline



Optical flow (Ch. 3)

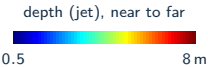
Unobserved: motion along an edge





Stereo depth (Ch. 4)

Unobserved: which points correspond



Data: MVSEC [35].

Dissertation: Methods: Chapter 2, Chapter 3, and Chapter 4; MVSEC [35].

Three estimators, each well posed

	Reconstruction (Ch. 2)	**Optical flow** (Ch. 3)	**Stereo depth** (Ch. 4)
Model supplies	the baseline, through the penalty	the motion along edges, through smoothness	the match, by encouraging two profiles to agree
Guarantee	unique and stable; one tridiagonal solve per pixel	unique when edges span two directions	unique inner solve at each candidate depth, with frozen weights

The common question: which quantities do the events determine, and which does the model supply?

Dissertation: Table 5.1; Theorem 2.2.5, Theorem 3.6.1, Theorem 4.6.3.

Events fix each pixel's brightness increments and leave one constant free

Observation at one pixel

Choose a unit-mass temporal kernel ϕ_{s_k}:

$$e(t) = \sum_{k \text{ at this pixel}} p_k \phi_{s_k}(t),$$

$$y_i = \int_{t_i}^{t_{i+1}} e(t)\, dt.$$

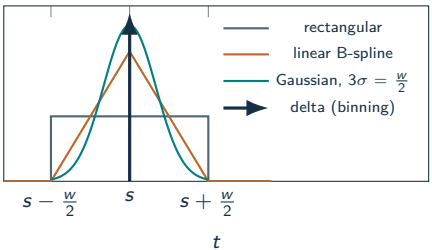
Increments of the unknown history

$$\mathbf{A} = \begin{pmatrix} -1 & 1 & & \\ & -1 & 1 & \\ & & \ddots & \ddots \end{pmatrix},$$

$$\mathbf{A}\mathbf{f} \approx \mathbf{y}, \qquad \ker \mathbf{A} = \operatorname{span}\{\mathbf{1}\}.$$

Dissertation: Eq. 2.4, Eq. 2.7, and Lemma 2.2.4.

Each kernel spreads an event's sign over nearby bins and conserves its total



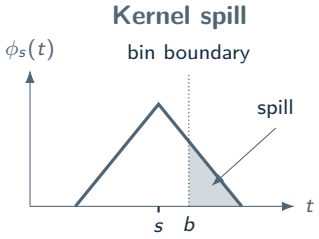
$$y_i = \sum_k p_k \int_{t_i}^{t_{i+1}} \phi_{s_k}(t)\, dt,$$

$$\sum_i y_i = \sum_k p_k$$

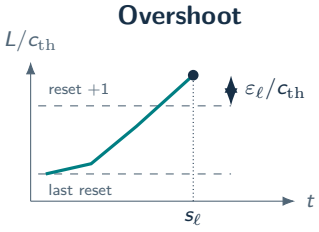
when every support lies inside $[t_0, t_N]$.

Dissertation: Proposition 2.2.1, Proposition 2.2.2; Table 2.2.

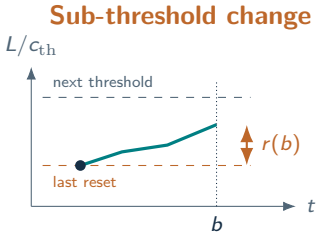
Three discrepancies connect event sums to brightness changes



Shaded kernel mass lies beyond b.



Event recorded above threshold.



Change since the last event.

Exactly, for any times $a < b$ at one pixel:

$$\frac{L(b) - L(a)}{c_{\mathrm{th}}} = y(a,b) - \underbrace{B(a,b)}_{\text{kernel spill at bin edges}} + \underbrace{\frac{1}{c_{\mathrm{th}}} \sum_{a < s_\ell \le b} \varepsilon_\ell}_{\text{overshoot}} + \underbrace{r(b) - r(a)}_{\text{sub-threshold change}} .$$

Dissertation: Physical connection (p. 24), Eq. 2.29, arbitrary-endpoint identity; threshold model [20]. Schematic diagram.

A unique, stable estimate for every positive weight

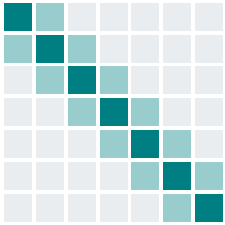
For fixed $\mathbf{\Lambda} = \mathrm{diag}(\lambda_i)$ with every $\lambda_i > 0$,

$$\mathbf{f}^* = \arg\min_{\mathbf{f}\in\mathbb{R}^n} \tfrac{1}{2}\|\mathbf{A}\mathbf{f} - \mathbf{y}\|^2 + \tfrac{1}{2}\|\mathbf{\Lambda}\mathbf{f}\|^2, \qquad \mathbf{K}_{\mathrm{time}} = \mathbf{A}^\top\mathbf{A} + \mathbf{\Lambda}^\top\mathbf{\Lambda} \succeq \lambda_{\mathrm{min}}^2\mathbf{I}.$$

Unique	$\mathbf{K}_{\mathrm{time}}\mathbf{f}^* = \mathbf{A}^\top\mathbf{y}$ has one solution	
Stable	$\|\delta\mathbf{f}^*\| \le (2/\lambda_{\mathrm{min}}^2)\,\|\delta\mathbf{y}\|$	one spurious event moves a history by at most $2/\lambda_{\mathrm{min}}^2$
Reference level	$\mathbf{1}^\top\mathbf{\Lambda}^2\mathbf{f}^* = 0$	quiet intervals carry the largest weights and set the level

Dissertation: Eq. 2.8, Theorem 2.2.5, Proposition 2.2.7; Tikhonov regularization [27].

Each pixel costs one tridiagonal sweep; streaming adds constant work per new frame



$$\mathbf{K}_{\text{time}} = \mathbf{A}^\top \mathbf{A} + \boldsymbol{\Lambda}^\top \boldsymbol{\Lambda}$$

diagonal $(1, 2, \ldots, 2, 1) + \lambda_i^2$
off-diagonals -1

$O(N)$ per pixel	Thomas sweep, no pivoting; tridiagonal and diagonally dominant
Independent pixels	a pixel without events returns $\mathbf{f} = 0$
Newest frame	the forward sweep alone gives it: constant work per pixel for each new interval

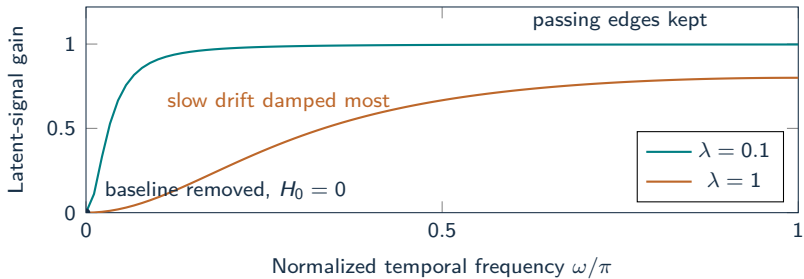
MVSEC indoor flying 1: dense solve 1.5 s → tridiagonal 0.03 s; agrees with MATLAB within 10^{-15}.

Dissertation: Eq. 2.7, Theorem 2.2.5; Thomas algorithm [32]; validation: edge implementation, October 2026.

The penalty removes the baseline and damps slow drift most; passing edges remain

For scalar $\mathbf{\Lambda} = \lambda \mathbf{I}$ and noiseless data $\mathbf{y} = \mathbf{A}\mathbf{f}_{\text{true}}$:

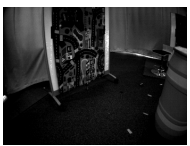
$$g_k = 4 \sin^2 \frac{\pi k}{2n}, \qquad H_k = \frac{g_k}{g_k + \lambda^2}, \qquad H_0 = 0.$$



Dissertation: Lemma 4.4.1, Theorem 4.6.21; cosine eigenbasis [81]. Curves: continuous extension.

Events alone recover scene structure with one convex solve and a unique global minimizer

MVSEC [35] indoor flying 2



Reference



Event only

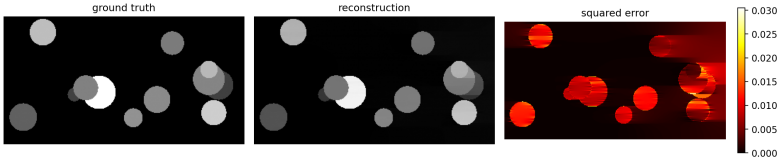
(PSNR in dB)	Bilevel [2]	Events
Inputs	events + frames	events
PSNR indoor	15.37	10.85
PSNR outdoor	8.41	8.71
Solve	outer + inner	one convex
Guarantee	existence; conditional local uniqueness	unique global minimizer

Dissertation: Table 2.8 and Table 2.9. Antil–Sayre [2], Sec. 3, Eqs. (3.1)–(3.3). Images: MVSEC indoor flying 2.

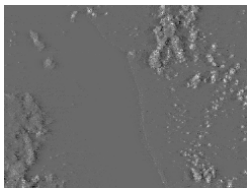
Known motion lets every view constrain one scene

Model $B_{\mathrm{SAR}} = (A \otimes I)\, P$ — θ scene · P known motion · A differences between views

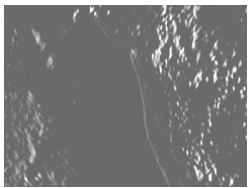
Simulated, known truth (v2e [46]); 1 px shift per view



Falcon Neuro on the ISS [38], near-nadir pass



each pixel on its own



all views, one scene (ESAR [45])

Dissertation: Figure 2.10, Eq. 2.47, Proposition 2.4.1; orbital data and reconstructions: ESAR [45], Falcon Neuro [38]; synthetic aperture [98].

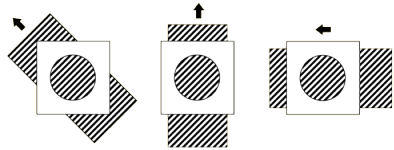
Three different motions look identical through a small window

Through the windows



The visible stripe motions look the same.

Reveal the full patterns



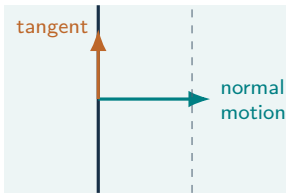
The patterns translate in different directions.

Publisher animations, unchanged: Wolfe et al. [109], Activity 8.2, Movies 5–6.

Dissertation: Proposition 3.3.4 and the following aperture discussion.

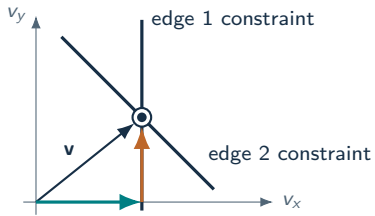
One edge fixes its normal speed; a second orientation fixes the rest

Image: one moving edge



dashed: the same edge a moment later;
events fire as it crosses each pixel

Velocity: what the events allow



normal: fixed by edge 1;
tangent: fixed by edge 2, if it shares the motion

Intersection of constraints [Adelson & Movshon 1982]. Dissertation: Proposition 3.3.4; requirement in Theorem 3.6.1.

Three representations, one quadratic variational problem

$$E_{\mathbf{B},b}(\mathbf{v}) = \underbrace{\frac{1}{2}\int_\Omega (\mathbf{v}^\top \mathbf{B}\mathbf{v} - 2b^\top \mathbf{v})}_{\text{directions the edges measure}} + \underbrace{\frac{\alpha^2}{2}\int_\Omega |\nabla \mathbf{v}|^2}_{\text{neighbors move alike}} \quad .$$

Representation	$\mathbf{B}$	b	Edge measurement from
Time surface	$\nabla\mathcal{T}\nabla\mathcal{T}^\top$	$\nabla\mathcal{T}$	crossing times
Voxel projection	$\nabla a \nabla a^\top$	$-a_s \nabla a$	transported event activity
Structure tensor	$\widetilde{\mathbf{J}}$	$-\mathbf{j}_t$	gradients pooled over a window

Dissertation: Theorem 3.6.1 and its three corollaries; data-plus-smoothness form [16].

Integrated directional information gives a unique flow in H^1

Unified well-posedness theorem

Let $\Omega \subset \mathbb{R}^2$ be bounded, connected, and Lipschitz; $\alpha > 0$. Assume

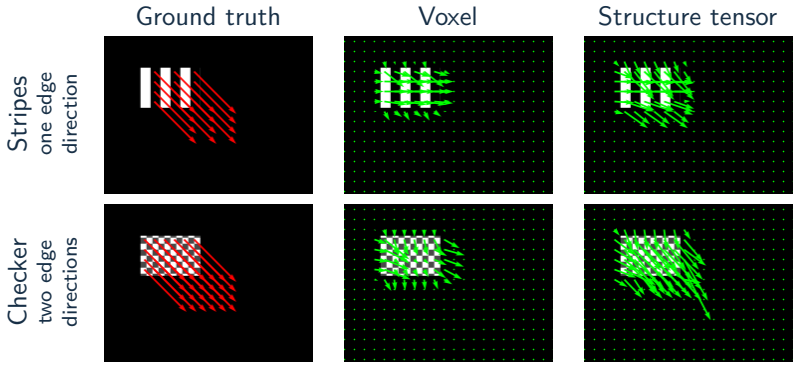
$$\mathbf{B} \in L^\infty, \quad \mathbf{B} = \mathbf{B}^\top \succeq 0 \text{ a.e.}, \quad b \in L^2, \qquad \boxed{\int_\Omega \mathbf{B}(\mathbf{x})\, d\mathbf{x} \succ 0}.$$

Then $E_{\mathbf{B},b}$ has a unique minimizer in $H^1(\Omega;\mathbb{R}^2)$.

Nonparallel edges control constant velocities; smoothness supplies motion where $\mathbf{B} = 0$.

Dissertation: Theorem 3.6.1, Eq. 3.59; classical quadratic-flow theory [57].

Two edge directions resolve more of the motion than one

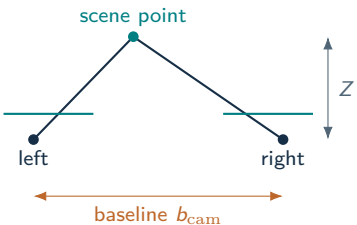


Dissertation: Figure 3.4; v2e [46] simulation, same window and arrow scale.

Stereo matches temporal profiles by encouraging agreement

$$\mathcal{J}_d(\mathbf{f}_L, \mathbf{f}_R) = \sum_{c \in \{L,R\}} \underbrace{\left(\tfrac{1}{2} \left\| \mathbf{A}\mathbf{f}_c - \mathbf{y}_c \right\|^2_{\mathbf{W}_c} + \tfrac{1}{2} \left\| \mathbf{\Lambda}\mathbf{f}_c \right\|^2 \right)}_{\text{each camera fits its own events}} + \underbrace{\tfrac{\gamma}{2} \left\| \mathbf{f}_L - \mathbf{f}_R \right\|^2}_{\text{encourages agreement}}$$

$\mathbf{y}_L$ at pixel (x, y), $\mathbf{y}_R$ at $(x - d, y)$; $\|\mathbf{v}\|^2_{\mathbf{W}} = \mathbf{v}^\top \mathbf{W}\mathbf{v}$, weights updated between solves



$$Z = \frac{f_{\text{cam}} b_{\text{cam}}}{d}, \qquad d = x_L - x_R > 0.$$

For fixed d and frozen weights, $\mathcal{J}_d$ has a unique minimizer.

Dissertation: Eq. 4.24, Section 4.5, Eq. 4.11, Theorem 4.6.3. Geometry is schematic.

WLS scores the most points, 2.66M, at competitive depth error

Method	Mean depth error (cm) ↓ FI1	FI2	FI3	FI4	log RMSE ×100 ↓	Scored points (M)
WLS (ours)	73.83	**58.35**	70.33	59.24	**44.54**	**2.66**
Piatkowska adaptation [78]	**67.96**	58.75	**68.88**	**54.63**	44.72	1.20



MVSEC [35] indoor flying 1–4; each method scored on its own estimates in $[0.5, 8]$ m.
Three more methods in the backup table.

Dissertation: Table A.5; MVSEC [35].

Each next step starts from a result in the dissertation

Sensor modeling

From: the error terms of slide 10

Measure them on real sensors; the stability bounds turn them into estimate bounds.

Learned weights

From: a smooth solution map

Learn $\mathbf{\Lambda}$ in an outer problem [110]; for $\mathbf{\Lambda} \succ 0$, $\mathbf{f}^*(\mathbf{\Lambda})$ is smooth.

Shared scenes

From: the null space of B_{SAR}

Find its motion-invariant modes for a trajectory (ESAR [45]); apply to orbital streams.

Edge hardware

From: the tridiagonal structure

Update only pixels with events; measure time and power on a Raspberry Pi 5 and edge sensors.

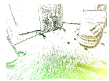
Dissertation: Chapter 5, Future Directions; physical connection (p. 24); Proposition 2.4.1. External paper cited above.

What the data constrain and what the models prove

Unique reconstruction and flow;
unique stereo inner solves.

Under the hypotheses summarized below.







	Reconstruction	**Optical flow**	**Stereo depth**
Events constrain	brightness increments	motion directions	paired temporal features
Model proves	**Unique, stable history** fixed positive weights	**Unique H^1 flow** integrated identifiability	**Unique inner solve** fixed disparity and weights

Stereo selects disparity by comparing candidates; accuracy also depends on the representation.

Dissertation: Theorem 2.2.5, Theorem 3.6.1, Theorem 4.6.3; Table 5.1.

Thank you.

Must every temporal penalty entry be positive?

For arbitrary $\mathbf{\Lambda}$, the exact algebraic criterion is

$$\mathbf{A}^\top \mathbf{A} + \mathbf{\Lambda}^\top \mathbf{\Lambda} \succ 0 \quad \Longleftrightarrow \quad \ker \mathbf{A} \cap \ker \mathbf{\Lambda} = \{0\}.$$

Why?

The quadratic form vanishes precisely when both $\mathbf{Az} = 0$ and $\mathbf{\Lambda z} = 0$.

On a connected first-difference chain, $\ker \mathbf{A} = \operatorname{span}\{\mathbf{1}\}$.

- A single nonzero diagonal penalty entry removes that constant mode.
- Fixing one reference value or imposing zero mean also removes it.

Takeaway. Any choice of reference level resolves the mathematical ambiguity; a measured reference would add physical information.

Dissertation: Lemma 2.2.4; discussion following Lemma 4.4.1.

Kernel performance depends on the scene and adaptive weights

	MVSEC indoor		MVSEC outdoor	
Kernel	PSNR	CN	PSNR	CN
B-spline	8.86	22.16	6.72	97.92
Delta	7.70	1375.75	7.97	53.34
Gaussian	8.84	20.22	6.81	98.90
Rectangular	8.24	423.59	7.20	148.33

- The temporal operator stays the same; represented data and adaptive penalties change.
- Conditioning and reconstruction bias are separate properties.
- Reference-image scores also measure the baseline the model selects.

Dissertation: Table 2.4, Table 2.5.

What the reconstruction scores measure

Quantity	What it compares	What it establishes
PSNR / SSIM	Reconstruction and reference images	Image agreement under the scoring convention
TC / TG	Temporal changes in the output	Smoothness of the returned sequence
ERC	Magnitudes of reconstructed changes and represented events	Correlation with event activity
CN	Sensitivity of a specified matrix	A numerical conditioning diagnostic

A simple check on interpretation

A constant sequence has $\mathrm{TC} = 1$ and $\mathrm{ERC} = 0$ under the stated conventions.

Takeaway. Consider reference fidelity, event consistency, and conditioning together. Smoothness alone rewards a static output.

Dissertation: Evaluation metrics (p. 31); Table 2.3; SSIM [40].

Stable estimates can still have regularization bias

For fixed scalar $\lambda > 0$, $\mathbf{y} = \mathbf{A}\mathbf{f}^\dagger + \boldsymbol{\eta}$, and $\mathbf{K} = \mathbf{A}^\top\mathbf{A} + \lambda^2\mathbf{I}$,

$$\mathbf{f}_\lambda - \mathbf{f}^\dagger = -\lambda^2\mathbf{K}^{-1}\mathbf{f}^\dagger + \mathbf{K}^{-1}\mathbf{A}^\top\boldsymbol{\eta}.$$

The two terms are regularization bias and propagated observation noise.

Takeaway. Stability controls sensitivity to the data; accuracy is a separate property.

Dissertation: Theorem 2.2.5, Proposition 2.2.7; direct bias-noise identity.

The physical event stream loses more than one baseline

The binned increment matrix has $\ker \mathbf{A} = \operatorname{span}\{\mathbf{1}\}$, but the ideal sensor can lose time-varying signals too.

$$L_1(t) = L_0, \qquad L_2(t) = L_0 + a\sin(2\pi t/T), \qquad 0 < a < c_{\mathrm{th}}, \quad t \in [0, T].$$

Both start at L_0, emit no events, and have positive intensities $I_j(t) = e^{L_j(t)}$.

Takeaway. The chain nullspace describes the discrete increment surrogate; the physical sensor adds sub-threshold ambiguity.

Dissertation: Eq. 2.1, Lemma 2.2.4; direct ideal-sensor example.

Frozen weights give a unique, stable inner solve

Fix d, $\mathbf{W}_L, \mathbf{W}_R \succ 0$, $\mathbf{\Lambda} \succ 0$, and $\gamma \geq 0$. Define

$$\mathbf{K}_c = \mathbf{A}^\top \mathbf{W}_c \mathbf{A} + \mathbf{\Lambda}^\top \mathbf{\Lambda}, \qquad \mathbf{M}_W = \begin{pmatrix} \mathbf{K}_L + \gamma \mathbf{I} & -\gamma \mathbf{I} \\ -\gamma \mathbf{I} & \mathbf{K}_R + \gamma \mathbf{I} \end{pmatrix}.$$

For $\mathbf{z} = (\mathbf{z}_L, \mathbf{z}_R)$,

$$\mathbf{z}^\top \mathbf{M}_W \mathbf{z} = \sum_{c=L,R} \left(\left\| \mathbf{W}_c^{1/2} \mathbf{A} \mathbf{z}_c \right\|^2 + \left\| \mathbf{\Lambda} \mathbf{z}_c \right\|^2 \right) + \gamma \left\| \mathbf{z}_L - \mathbf{z}_R \right\|^2.$$

Conclusion

$\mathbf{M}_W \succeq \lambda_{\min}^2 \mathbf{I} \succ 0$: the frozen quadratic has exactly one global minimizer.

Dissertation: Eq. 4.34, Theorem 4.6.3, Theorem 4.6.11.

Motion-induced events and the aperture problem

For a straight edge under steady illumination, let $\mathbf{n}$ be its unit normal, $\mathbf{t}_\perp$ a unit tangent, and s its normal speed:

$$L(\mathbf{x}, t) = \ell(\mathbf{n} \cdot \mathbf{x} - st), \qquad \nabla L = \ell' \mathbf{n}, \qquad L_t = -s\ell'.$$

Where $\ell' \neq 0$, brightness constancy gives

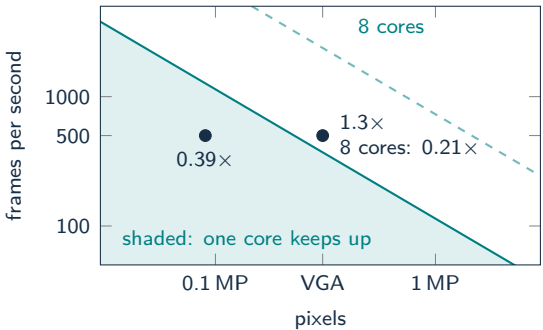
$$L_t + \nabla L \cdot \mathbf{v} = 0 \iff \mathbf{n} \cdot \mathbf{v} = s \iff \boxed{\mathbf{v} = s\mathbf{n} + \tau \mathbf{t}_\perp, \quad \tau \in \mathbb{R}.}$$

- Every such constant velocity gives the same local pattern and ideal events from the same reset reference.
- For $s = 0$, pixel values stay constant in the ideal edge interior; endpoints lie outside the observation window.

Takeaway. The edge determines normal speed; its tangential velocity remains free. Illumination changes can also trigger events under the sensor threshold rule.

Dissertation: Proposition 3.3.4 and following aperture remark; Eq. 2.1.

One CPU core streams about 114 million pixel updates per second, with 16 bytes of state per pixel



$$\text{pixels} \times \text{frames/s} \;\leq\; 1.14 \times 10^8 \text{ per core}$$

8.8 ns per pixel update; 16 B of state per pixel

Batch work follows scene activity: skipping pixels without events cuts it 10× at 6.8% active (0.43 → 0.04)

Points: two recordings, a frame every 2 ms, float32; labels give processing time ÷ recording time. 8-core line: measured at VGA.

Dissertation: Timings: edge implementation, October 2026; Thomas algorithm [32].

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